

Laboratory work 3

Programming of cyclic algorithms

Objective : To acquire skills in using loop operators, learn to program computational processes with a known number of repetitions, learn to program algorithms of an iterative cyclic structure.

Short theoretical information

A cycle (repetition) is a process of performing a certain set commands a certain number of times. The C language contains three repetition operators:

for, while, do while.

3.1. A loop statement with a parameter

The loop operator with a parameter has the form:

```
for (<Expression 1>; <Boolean Expression 2>; <Expression 3>)
{
    <Cycle Body>
}
```

Expression 1 is intended to prepare the loop and is executed alone times (loop initialization). Mostly, the initial values of the cycle variables are set here (the cycle is prepared). In logical expression 2, the condition for exiting the loop is written. In expression 3 - commands for changing cycle parameters.

In the Body of the loop block, write a block of commands that will be repeated in the loop.

The for loop is executed if the Boolean expression is 2 (condition) takes the value TRUE. If it takes the value FALSE at least once, the program exits the loop and executes the next statement that follows the body of the for loop.

If with the help of one of the expressions it is necessary to perform several actions, then use the "coma" command. Expressions 1 and 3 or one of them in the for command may be missing. In this case, omit the symbol “;” can not.

The value of a variable should not be substituted inside the loop body loop, although such a change is not recorded as an error and does not affect the number of executions of the loop body.

After the operation of the statement of the for loop of the value of the loop variable is equal to the final value. If the loop body is not executed, then the value of the variable is undefined.

Example 3.1. Display lowercase letters of the Latin alphabet in a different order.

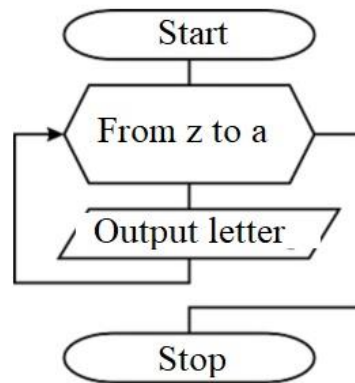


Fig. 3.1 - Block diagram of the algorithm of the program

Program text

```

#include <stdio.h>
void main()
{
for (char i = 'z'; i >= 'a'; i--)
printf("%c", i);
}
Results of the program
zyxwvutsrqponmlkjihgfedcba

```

Example 3.2.

Find the element of a geometric progression, knowing the first element and the number of the element.

A geometric progression is a sequence, each term of which, starting from the second, is equal to the previous one multiplied by the same number: $b_1, b_1q, b_1q^2, b_1q^3, \dots$. The number q is the denominator of geometric progression. The number b_1 is the first term of the geometric progression. The first term and the denominator q of the geometric progression can be any numbers. Formula of the n th term of the geometric progression: $b_n = b_1q^{n-1}$.

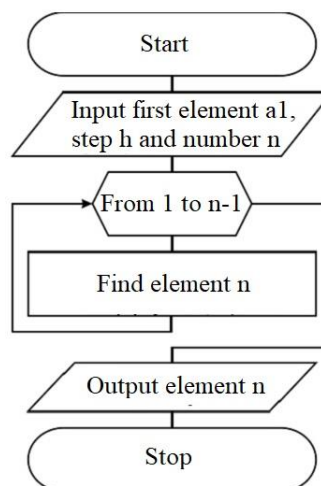


Fig. 3.2 - Block diagram of the algorithm of the program

Program text

```
##define CRT_SECURE_NO_WARNINGS
#include <stdio.h>
int main()
{
double firstItem, step;
int index;
printf("Input first
item of geometric progression\n");
scanf("%lf",
&firstItem);
printf("Input step of
geometric progression\n");
scanf("%lf", &step);
printf("Input index of
element of geometric
progression\n");
scanf("%d", &index);
for (int i = 1; i < index; i++)
{
firstItem *= step;
}
printf("Item with index
%d equal %.2f", index, firstItem);
return 0;
}
```

Results of the program:

Input first item of geometric progression

3

Input step of geometric progression

5

Input index of element of geometric progression

2

Item with index 2 equal 15.00

Tasks of laboratory work**Task 1**

Tabulate the function $y = f(x)$ on the interval $[a; b]$ with a step $h = (b - a)/N$, where N is an integer (entered from the keyboard).

The program should provide for the calculation of the values specified in the tasks (Table 3.2).

Table 3.2

Variant	Function $y = f(x)$	Interval	
		a	b
1	$y = x^2 + 16/x - 12$	1	4
2	$y = 2\arctg x$	0	2
3	$y = 6 + \sin^2 x$	0	4
4	$y = x(5 + x)$	1	6
5	$y = x^{4x}$	0	6

Task 2

Develop a program to find the function y for a given x . Set the value of x independently, given that x is real (not an integer). Calculate the function y for five different values of x .

Task options:

- $y = \frac{(x-2)(x-4)(x-6)\dots(x-32)}{(x-1)(x-3)(x-5)\dots(x-31)};$
- $y = \frac{(x+2)(x+4)(x+6)\dots(x+32)}{(x-1)(x-5)(x-9)\dots(x-1024)};$
- $y = \frac{(x-1)(x-3)(x-6)\dots(x-243)}{(x+1)(x+3)(x+5)\dots(x+31)};$
- $y = \frac{(x-1024)(x-1000)(x-996)\dots(x-2)}{(x-1)(x-3)(x-5)\dots(x-31)};$
- $y = \frac{(x+2)(x-4)(x+6)\dots(x+32)}{(x-1)(x+3)(x-5)\dots(x-31)}.$

Task 3.

Write a program according to the option.

- A given sequence of N integers. Determine how much of positive numbers occurs up to the first negative number.
- A given sequence of N integers, among which there are 0. Determine how many numbers up to 0 occur.
- Given a sequence of arbitrary natural numbers followed by 0. Find the sum of these numbers.
- Given an integer N . Find the sum of the digits of this number.
- Given an integer $N > 0$. Find the largest integer for which the condition $2^k < N$ is fulfilled.

Task 4.

Tabulate the function $y = f(x)$ on the interval $[a; b]$ with a step $h = (b - a)/N$, where N is an integer (entered from the keyboard). The program provides for the calculation of the values indicated in the data (Table 3.3).

Table 3.3

Variant	$y = f(x)$	Interval		Calculated values
		a	b	
1	$y = x^2 + 16/x - 16$	1	4	Arithmetic average of all y
2	$y = \operatorname{arctg} x$	0	3	Amount and quantity $0,71 < y < 2$
3	$y = 4 - x - 4/x^2$	1	4	Quantity $y < 0,5$ i $y > -0,5$
4	$y = \operatorname{arctg} x$	0	2	Arithmetic average $0 < y < 0,5$
5	$y = \sin^3 x$	0	3	Amount $y > 1$ and quantity $y < 1$

Task 5.

Calculate the sum of the series $\sum_n^\infty k a_n$ with accuracy $\varepsilon = 0.0001$. According to the compiled program, perform calculations at the given value of x and output the number of terms that were summed up.

Table 3.4

Variant	a_n	k	x
1	$(-1)^n \frac{\cos(nx)}{x^2}$	1	1,25
2	$(-1)^n \frac{(1+x)^{2n}}{n}$	1	1,34
3	$(-1)^{n+1} \frac{x^{2n}}{2n(2n-1)}$	0	0,45
4	$(-1)^{n-1} \frac{x^{2n-1}}{2n-1}$	0	0,65
5	$\frac{n}{(2n+1)} x^{n-1}$	1	0,91